A Development cases

Taipei chargers
Maximum coverage; two scales

YouBike
Closed tour; two scales

Taipei AED
Balanced partition; two scales

London ambulance
p-median; two scales

Berlin playgrounds
Incremental coverage; two scales

Fresh runs; no saved model or cache

B Mechanism ladder

1. **Direct LLM**
   Whole-system baseline
2. **Structured workflow**
   Typed interfaces + toolkit
3. **Workflow with checkers**
   Add separate computations
4. **Complete workflow**
   Add cross-stage repair
5. **Workflow with retrieval**
   Add modelling patterns

Without clarification web lookup
Acquisition still searches the web

C External judging

Truth withheld from agents

1. **Data correct**
   Entities, coordinates, values
2. **Model correct**
   Feasibility + objective ranking
3. **Usable delivery**
   Requirements + reference quality

Joint success = all three
Three repetitions planned

Protocol, not completed results. Freeze code, prompts, cases, judge and truth; direct tools differ.